

SURP 2025

Ex-ante Cybersecurity Risk Research

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Week 0

LinkedIn Learning



Python Essentials Training:

- Gained knowledge on basic operators, if/while/for statements, classes, functions, read/write/open files, etc.

Introduction to Data Science:

- Learned the differences between computational tools (python and r) * Learned data visualization for both quantitative/qualitative data
- Learned how to reduce bias in probability sampling

Pandas Essential Training:

- Understood the fundamentals of pandas including lists, dictionaries, series, filtering, and dataframes

Week 1

Constructed Python Database

- This data base was intended to scrape through 10-K filings from different companies and return the following items:
 - Full 10-K as readable text
 - Item 1A – SEC guidelines in 2023 * Item 1C – SEC guidelines in 2018

Details:

- Converted the 10K XML format to human-readable text
- Used different packages to create a more efficient coding process including:
 - BeautifulSoup
 - SEC-Edgar tools - Filtered by year (e.g., 2025)

Outcome:

- This week provided me with foundational knowledge on how future coding will be approached and allowed me to gain hands-on experience in understanding the complexity involved in developing efficient code for this research project. —

Week 2

Extracted information from 10-K filings based on keywords and questions

- This week, I successfully extracted data from SEC Filings using EDGAR Full Text Search. This gave me valuable information regarding different cybersecurity topics from all filings since 2001 through the use of the keywords below.

Keywords:

- Cyber Insurance, Cyber Insurance Coverage, Cyber Liability Insurance, Cybersecurity Incident Policy, Cybersecurity Insurance, Cyber Risk Insurance, Network Security and Private Liability Insurance, Data Breach Insurance, Privacy Liability Insurance, Technology Errors and Omissions, Cyber Extortion Coverage, Ransomware Insurance, Incident Response Coverage

Answered the following questions:

- What does the insurance cover?
- What are the coverage limits?
- What are the coverage premiums?
- What recoveries are available?
- Are there any additional notes?

Week 3

Project Shift

- This week, we were working on creating a RAG on each 10-K and using OpenAI embedding models to measure precision in giving AI models full text or RAG
 - Focused on using LANG chains, Turning Chunks into vectors, Sending batches to OpenAI

However, difficulties struck in:

- Fully understanding each complex topic enough to program and explain each concept
- Time consumption in creating these models

Because of this

- We decided to shift gears and create a more efficient way to conduct our research

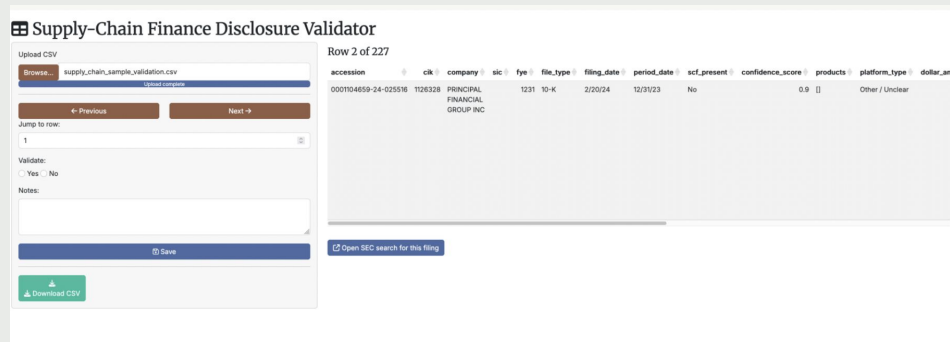
Week 4

SEC Filing Research

- This week, I compared AI-extracted passages with the original SEC filing shown in EDGAR using:
 - An application in R and Rstudio
 - Uploaded CSV file containing AI outputs

Topics I looked for:

- **Coverage:** Was any important information dropped?
- **Fidelity:** No new language introduced
- **Context:** Anything quoted out of context or anything misleading
- **Confidence Score:** Anything under a 0.7 confidence score was flagged



The screenshot shows the 'Supply-Chain Finance Disclosure Validator' web application. On the left, there is a sidebar with an 'Upload CSV' section containing a file browser and a 'Upload CSV' button. Below this are navigation buttons for 'Previous' and 'Next', a 'Jump to row' input field set to '1', a 'Validate' section with radio buttons for 'Yes' and 'No', a 'Notes' text area, a 'Save' button, and a 'Download CSV' button. The main area displays 'Row 2 of 227' and a table of data.

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0001104659-24-025516	1126328	PRINCIPAL FINANCIAL GROUP INC	1231	10-K	2/20/24	12/31/23	No	0.9		Other / Unclear		

Week 5

FDIC Comments/Letters Research

- This week, I split a 795 page document containing various archived letters between companies (primarily banks) and the FDIC. I then parsed through every letter in this document and extracted important information regarding crypto-activities.

Different Categories of Important Information

- Date
- Type of Activity proposed
- Channel: Assets or Technology or “Assets & Technology”
- FDIC Stance
- Any additional comments

Important Information learned about general research:

- Research must be done in a certain way in order for it to be used efficiently as I conducted the same parsing twice due to formatting purposes

Week 6-7

Conducted Prompt Engineering through validating AI information

- This week, I validated AI information that was parsed from companies' Item 1As and Item 1Cs. While there were several different blocks that was given, we focused primarily on the investment block regarding cybersecurity
 - Batch Prompting (Open AI) → structured JSON outputs
 - Parsing + Scoring Pipeline and the exact fields

Key Findings:

- There was a lot of redundancy in findings and fixings for AI including:
 - Mistaking the opex
 - Training
 - And external assurance

Learned the following:

- AI prompts should be tinkered to gain more effective output
- Much of these company filings are vague in listing cybersecurity information

Week 8-10

Create, collaborate, and improve Cyber-Risk Extraction Prompt

- Over the following weeks, I worked with my colleague in order to create the most optimal prompt for our constructed LLM
 - Improved grading rubric-based prompt on certain filing fields
 - Cross-checked findings to ensure accuracy

Key Findings:

- We improved the scoring rubric so the model could better assess company cybersecurity disclosures in categories such as:
 - Cyber-based investments
 - Cyber-insurance
 - Past cyber-related incidents
 - Cyber-based mitigation strategies

Learned the following:

- The more relevant and specific information included in the prompt, the stronger and more accurate the model's assessment results